

MH2500 Probability and Introduction to Statistics

Slides for Week 4-2

Topics: Binomial Distribution and Poisson Distribution

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Bernoulli Random Variable

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Example: The indicator variable of event A is a Bernoulli variable with $p = P(A)$.

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$$p(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

$$\sum_{k=0}^n p(k) = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} = (p + (1 - p))^n = 1.$$

Recall the Example

Toss a fair coin three times and let X be the number of heads.

It is a **binomial random variable** with parameters 3 and $\frac{1}{2}$:

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$$P\{X = 0\} = \binom{3}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$P\{X = 1\} = \binom{3}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2 = \frac{3}{8}$$

$$P\{X = 2\} = \binom{3}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 = \frac{3}{8}$$

$$P\{X = 3\} = \binom{3}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 = \frac{1}{8}$$

Example

A player bets on one of the numbers 1 through 6. Three dice are rolled and if the number bet by the player appears i times ($i = 1, 2, 3$), the player wins i units; if the number does not appear then the player loses 1 unit.

Is this game fair to the player?

Let X denote the number of occurrences of the number bet.

Then $X \sim \text{Bin}(3, 1/6)$ and

$$P\{X = 0\} = \binom{3}{0} \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = \frac{125}{216}$$

$$P\{X = 1\} = \binom{3}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = \frac{75}{216}$$

$$P\{X = 2\} = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = \frac{15}{216}$$

$$P\{X = 3\} = \binom{3}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = \frac{1}{216}$$

Let Y denote the winning of the player. We have:

$$P\{Y = -1\} = P\{X = 0\} = \frac{125}{216}$$

$$P\{Y = 1\} = P\{X = 1\} = \frac{75}{216}$$

$$P\{Y = 2\} = P\{X = 2\} = \frac{15}{216}$$

$$P\{Y = 3\} = P\{X = 3\} = \frac{1}{216}$$

Thus,

$$\mathbb{E}(Y) = (-1) \cdot \frac{125}{216} + \frac{75}{216} + 2 \cdot \frac{15}{216} + 3 \cdot \frac{1}{216} = -\frac{17}{216} < 0.$$

The game is not fair to the player.

Question

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Recall the identity

$$\binom{n}{i} = \frac{n}{i} \cdot \binom{n-1}{i-1}.$$

Express $\mathbb{E}(X^k)$ in lower-order moments. We will have more examples in the tutorial.

$$\begin{aligned}\mathbb{E}(X^k) &= \sum_{i=0}^n i^k \binom{n}{i} p^i (1-p)^{n-i} \\&= \sum_{i=1}^n i^k \frac{n}{i} \binom{n-1}{i-1} p^i (1-p)^{n-i} \\&= n \sum_{i=1}^n i^{k-1} \binom{n-1}{i-1} p^i (1-p)^{n-i} \\&= np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} \\&= np \mathbb{E}((Y+1)^{k-1})\end{aligned}$$

where

$$Y \sim \text{Bin}(n-1, p).$$

Proposition

Suppose that $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(n - 1, p)$. Then for $k \geq 1$,

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$$\mathbb{E}(X) = np \mathbb{E}((Y + 1)^0) = np.$$

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E} X)^2 \\ &= np \mathbb{E}(Y + 1) - n^2 p^2 \\ &= np[(\mathbb{E} Y) + 1] - n^2 p^2 \\ &= np[(n - 1)p + 1] - n^2 p^2 = np(1 - p)\end{aligned}$$

Properties of Binomial Random Variables

Let $X \sim \text{Bin}(n, p)$ for $p \in (0, 1)$. As k goes from 0 to n , $P\{X = k\}$ first increases and then decreases, reaching its largest value when $k = \lfloor (n + 1)p \rfloor$.

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Proof:

$$\frac{P\{X = k\}}{P\{X = k - 1\}} = \frac{\binom{n}{k} p^k (1-p)^{n-k}}{\binom{n}{k-1} p^{k-1} (1-p)^{n-k+1}} = \frac{n-k+1}{k} \cdot \frac{p}{1-p}.$$

$$\begin{aligned} P\{X = k\} \geq P\{X = k - 1\} &\Leftrightarrow p(n - k + 1) \geq (1 - p)k \\ &\Leftrightarrow k \leq (n + 1)p \end{aligned}$$

This shows that $P\{X = k\}$ first increases and then decreases, reaching the largest value when $k = \lfloor (n+1)p \rfloor$.

Poisson Distribution

The **Poisson frequency function** with parameter λ ($\lambda > 0$) is

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

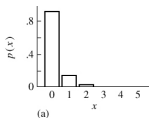
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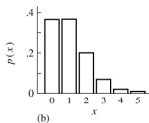
The frequency function sums to 1 since

$$\sum_{k=0}^{\infty} P(X = k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1$$

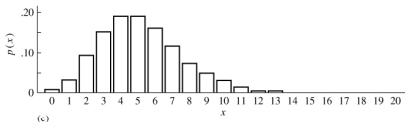


(a) $\lambda = .1$

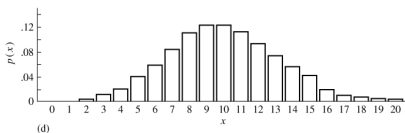
Figure 2.6
Poisson frequency
functions.



(b) $\lambda = 1$



(c) $\lambda = 5$



(d) $\lambda = 10$

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See also [http:](http://demonstrations.wolfram.com/PoissonDistribution/)

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The Poisson distribution can be derived as the limit of a binomial distribution with $np = \lambda$, as $n \rightarrow \infty$.

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Proof: For a fixed n , we have $p = \frac{\lambda}{n}$ and so

$$\begin{aligned} p(k) &= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\ &= \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \frac{n!}{(n-k)!} \frac{1}{n^k} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}. \end{aligned}$$

As $n \rightarrow \infty$,

$$\frac{n!}{(n-k)!} \frac{1}{n^k} \rightarrow 1$$

$$\left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}$$

$$\left(1 - \frac{\lambda}{n}\right)^{-k} \rightarrow 1.$$

Thus $p(k) = (\lambda^k/k!) \cdot (\text{three factors above}) \rightarrow \lambda^k e^{-\lambda}/k!$.

Poisson Distribution (continued)

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Consider the number of calls arriving at a telephone exchange in one hour. We assume that

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Number of calls X made in one hour is usually modelled following Poisson distribution

Example

Suppose that an office receives telephone calls as a Poisson process with a rate of 0.5 calls per min. Find

- (i) The probability that there are no calls in a 5 min interval
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- (i) The probability of no calls in a 5 min interval is

$$\frac{2.5^0}{0!} e^{-2.5} = e^{-2.5} = 0.082.$$

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- (ii) The probability of one call in a 5 min interval is

$$\frac{2.5^1}{1!} e^{-2.5} = 2.5e^{-2.5} = 0.205.$$

Question

Let $X \sim \text{Poisson}(\lambda)$. What are $\mathbb{E}(X)$ and $\text{Var}(X)$?

$$\begin{aligned}\mathbb{E}(X) &= \sum_{i=1}^{\infty} \frac{i \cdot \lambda^i}{i!} e^{-\lambda} \\&= \lambda \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} e^{-\lambda} \\&= \lambda \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} e^{-\lambda} \\&= \lambda\end{aligned}$$

$$\begin{aligned}
\mathbb{E}(X^2) &= \sum_{i=1}^{\infty} \frac{i^2 \cdot \lambda^i}{i!} e^{-\lambda} \\
&= \lambda \sum_{i=1}^{\infty} \frac{i \lambda^{i-1}}{(i-1)!} e^{-\lambda} \\
&= \lambda \sum_{j=0}^{\infty} \frac{(j+1) \lambda^j}{j!} e^{-\lambda} = \lambda \sum_{j=0}^{\infty} \left[\frac{j \lambda^j}{j!} e^{-\lambda} + \frac{\lambda^j}{j!} e^{-\lambda} \right] \\
&= \lambda \left[\sum_{j=0}^{\infty} \frac{j \lambda^j}{j!} e^{-\lambda} + \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} e^{-\lambda} \right] \\
&= \lambda(\lambda + 1)
\end{aligned}$$

Hence

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \lambda(\lambda + 1) - \lambda^2 = \lambda.$$

$$\text{Bin}(n, p) \rightarrow \text{Poisson}(\lambda), \quad np = \lambda, n \rightarrow \infty$$

$$\text{Expectation} \quad np \rightarrow \lambda$$

$$\text{Variance} \quad np(1 - p) \rightarrow \lambda$$